Quantum Glitchy Period Finding

Status report and open problems

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Compiled October 2026 from the 2019 poster, simulation code and logs in this repository

Abstract. Quantum period finding assumes an exactly periodic oracle. We study a periodic sequence corrupted by k insertions or deletions ("glitches"). Each glitch shifts every later entry, so the corrupted function disagrees with every periodic function on a constant fraction of its domain. This note records the state of the 2019 project: the idealized model used throughout, the exact measurement distribution, the one-glitch theorem (probability near a good outcome at least 0.447 times ideal), the multi-glitch conjecture, and the numerical evidence. The 2019 materials (poster, code, logs) contain no written proofs. Material marked new in this note was reconstructed or added while compiling this note. It includes an identification of the one-glitch constant as $8/(8+\pi^2)$, a Parseval reformulation of the multi-glitch problem, a crude bound with quadratic radius, and a family of configurations showing that the conjectured constant $1/2$ would be asymptotically sharp. Separately, an exact residue-counting argument proves total probability at least $1/(2k+1)$ at the ideal outcomes themselves after any k literal glitches, with a stronger bound approaching $1/(k+1)$ when $k = o(N)$. A step-by-step proof and a worked example explain this total-mass guarantee; it does not resolve the stronger per-outcome conjecture or establish period recovery. A separate Fejér-kernel argument proves total probability at least $1/3$ within distance k of the rounded peaks for every $k \geq 1$, even when $r \nmid N$. A list of open problems closes the note.

How to read the status labels. from 2019 work the statement or formula appears in the 2019 poster, code or logs. 2019 claim, proof not on file stated on the 2019 poster; no proof is in the repository. new in this note added while compiling this note. proof below a complete argument is given here. Lean checked the general-k arguments in Section 4.1 and Section 4.2 have checked Lean proofs. proof sketch only the argument is outlined and the details still need to be checked. numerical evidence only supported by computation only. conjecture open. Statements marked new in this note were checked numerically with `uv run glitchy.py check` (Appendix A). They were prepared with AI assistance and have not been independently refereed.

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1 Summary of the current state

- **Model.** All 2019 analysis and simulation use one idealized *block-shift model* (Definition 1). In it every glitch sits on a period boundary and shifts all later copies of the period by ± 1. Its measurement distribution has an exact closed form (Lemma 2).
- **One glitch.** The poster states that for $r \mid N$ the probability of an outcome within distance 1 of a good outcome is more than 0.447 times the ideal probability. In the continuum limit we reconstruct the argument (Theorem 10). The neighbourhood ratio is at least $1 - 4u(1-u) + (8/\pi^2)\sin^2(\pi u)$, and Jordan's inequality turns this into $8/(8+\pi^2) = 0.4477...$, matching the poster's constant. The sharp value is 0.6400.
- **Several glitches.** The poster's "gruesome expression" is exactly the continuum neighbourhood mass (Section 4.4). The 2019 target was to prove that it is at least $1/2$ (Conjecture 7). Exhaustive simulations for $k = 2, 3, 4$ give worst cases 0.772, 0.614, 0.689 (Table 2). The 2019 grid search for $k = 4$ had found only 0.723. The worst good outcome is $c = r/2$ for $k = 1, 3$ but not for $k = 2, 4$.
- **Added in this note.** (a) The neighbourhood mass is 1 minus the high-frequency energy of a unimodular step function, with an explicit formula in terms of jumps (Proposition 8). (b) Radius $L = \lceil 4(k+1)^2/\pi^2 \rceil$ already gives mass $\geq 1/2$ (Corollary 9), so the open problem is to reduce the radius to $O(k)$. (c) Error bounds for passing between the discrete and continuum models (Lemma 6, Proposition 12). (d) A packed family whose mass at radius k decreases towards $1/2$, reaching 0.521 at $k = 200$ (Section 8.1). (e) Any bound uniform in c needs radius at least $(k+1)/2$ (Proposition 11).
- **Any number of glitches, total mass.** For k literal insertions/deletions at arbitrary positions, with arbitrary inserted symbols, the total probability at the exact ideal outcomes is at least $r/((k+1)r + I) \geq 1/(2k+1)$, where $I \leq k$ is the number of surviving inserted entries (Theorem 3). A second bound is $(1 - I/N)^2/(k+1)$. Normalized Model I gives $1/(k+1)$ (Proposition 4). These rates are sharp up to constants on the exact set. Within distance k, the constant $133/358$ of Theorem 5 also holds when $r \mid N$. All of these are averaged statements, not lower bounds near each good outcome.
- **Nondivisible periods, total mass.** For every $k \geq 1$, literal edits have total probability at least $1/3$ within distance k of the rounded peaks, whether or not $r \mid N$ (Theorem 5). This is again a union-of-windows guarantee, not a per-outcome bound.
- **Period recovery, numerically.** The exact success probability of the recovery procedure stays at least 0.81 times its glitch-free value for the worst placements found, with $k \leq 4$ and $N \leq 3001$ (Table 3). The true period provably passes the classical test once $\tau > (kr + A)/(N - r)$.
- **Not done.** No multi-glitch theorem uniform in c at radius $O(k)$. No period-recovery theorem (Section 9). No per-outcome analysis of $r \nmid N$. No public follow-up paper or manuscript exists as of October 2026 (Section 10, item 8).

2 Standard period finding

Let $f : \mathbb{Z}_N \to W$ have period r, with $r < \sqrt{N}$. For most of this note we also assume[1]

$$r \mid N, \quad M := N/r, \quad f(x) = w_{x \bmod r} \text{ with } w_0, ..., w_{r-1} \text{ distinct.} \tag{1}$$

The algorithm prepares $N^{-1/2} \sum_x |x\rangle|f(x)\rangle$, applies the quantum Fourier transform to the first register and measures. Outcome y appears with probability

[1]The poster's figure 5 treats $r \nmid N$; nothing is proved there.

$$P_f(y) = \frac{1}{N^2} \sum_{u \in W} \left| \sum_{x : f(x) = u} e^{2\pi i x y / N} \right|^2 . \tag{2}$$

For the ideal function, value w_a is taken on $\{a + mr : 0 \le m < M\}$, and the inner sum is $e^{2\pi i a y/N} \sum_{m=0}^{M-1} e^{2\pi i m y/M} = M e^{2\pi i a y/N} \mathbb{1}[M \mid y]$. Hence

$$P_0(y) = \frac{1}{r} \cdot \mathbb{1}[M \mid y], \quad G := \{cM : 0 \le c < r\}. \tag{3}$$

From a good outcome $y = cM$ we get $y/N = c/r$. Reducing the fraction gives $r/\gcd(c, r)$, and $\Pr[\gcd(c, r) = 1] = \varphi(r)/r = \Omega(1/\log\log r)$. When $r \nmid N$, the peaks sit at the integers nearest cN/r. Continued fractions then recover c/r because $r^2 \le N$. This is where the assumption $r < \sqrt{N}$ is used.

3 The glitch model

3.1 Informal picture

An *insertion* adds one symbol to the periodic sequence and shifts everything after it by one place. A *deletion* removes one symbol and shifts everything after it back by one place. Figure 1 shows both. With period 3, `abcabcabc…` becomes `abcadbcabc…` after an insertion and `abcacabc…` after a deletion. After the glitch the sequence is still locally periodic, but its phase has moved. A single glitch therefore changes f on about half of the domain, so perturbation bounds that count changed entries are useless (research prompt, Approach 2).

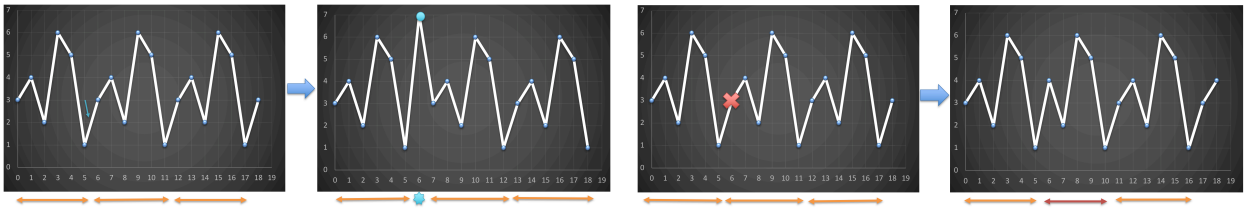


Figure 1: Poster figures 1–2. A periodic function before and after one insertion (left pair) and one deletion (right pair).

3.2 The block-shift model used in 2019

Definition 1 (Block-shift model, "Model I"). from 2019 work Assume Eq. (1). A *k-glitch configuration* consists of

- block boundaries $0 = m_0 \le m_1 \le \ldots \le m_k \le m_{k+1} = M$. Glitch j sits between copy $m_j - 1$ and copy m_j of the period;
- signs $\varepsilon_1, \ldots, \varepsilon_k \in \{+1, -1\}$, where $+1$ is an insertion and -1 a deletion.

Put $s_1 = 0$, $s_{j+1} = s_j + \varepsilon_j$ (cumulative shift) and $B_j = \{m_{j-1}, \ldots, m_j - 1\}$ for $j = 1, \ldots, k+1$. Copy $m \in B_j$ of the period starts at $mr + s_j$, so value w_a occupies $X_a = \{a + mr + s_j : 1 \le j \le k+1, m \in B_j\}$, read modulo N. The Model I distribution is

$$P_k(y) = \frac{1}{N^2} \sum_{a=0}^{r-1} \left| \sum_{x \in X_a} e^{2\pi i x y / N} \right|^2 = \frac{r}{N^2} \left| \sum_{j=1}^{k+1} e^{2\pi i y s_j / N} \sum_{m \in B_j} e^{2\pi i m y / M} \right|^2 . \tag{4}$$

The second equality holds because a enters only through the unimodular factor $e^{2\pi i a y/N}$.

Remark. **Correspondence with the code.** Eq. (4) is `distribution` in `glitchy.py`, which is `probability_distr_with_insertion_deletion` from the 2019 scripts. There `ListK` $= (m_0, ..., m_{k+1})$, `Shift` $= (s_1, ..., s_{k+1})$, and `abs(z)**2/(mb*N)` equals $r|z|^2/N^2$. Every 2019 run used equal signs, so $s_j = \pm(j-1)$: the code picked "all insertions" or "all deletions" at random. For $k < r$ the points of each X_a are distinct modulo N, so $\sum_y P_k(y) = 1$, which `glitchy.py check` confirms.

Remark. **What Model I idealizes.** Take the *literal model*: an infinite r-periodic sequence, k edits at arbitrary positions, arbitrary inserted symbols, and keep the first N symbols. Model I differs from it in three ways. (i) A glitch strictly inside a period, between residues $b-1$ and b, moves only the classes $a \geq b$ of that copy. Model I moves the whole copy. (ii) The inserted symbols are dropped. (iii) Model I keeps exactly M copies of every value and wraps positions outside $[0, N)$ around. The literal model truncates instead. Each effect changes at most $O(k)$ points of each class. Proposition 12 bounds the resulting error. The 2019 materials do not settle several questions raised in the research prompt. The bounds are meant to hold for every placement: the poster's conjecture is worst-case, and the code enumerated placements exhaustively or at random. Inserted values play no role in Model I. Adjacent glitches are allowed ($m_j = m_{j+1}$ gives an empty block), but two glitches inside one period are not modelled. k counts operations, and the net displacement is $|s_{k+1}| \leq k$. Mixed sign patterns are allowed by Eq. (4). The 2019 runs did not simulate them; see Section 7.3.

4 The distribution near good outcomes

Lemma 2 (Exact form near good outcomes). from 2019 work proof below Write $y = cM + h$ with $0 \leq c < r$ and $|h| < M/2$, and let $\omega = e^{2\pi i/r}$. Then

$$R_c(h) := \frac{P_k(cM+h)}{P_0(cM)} = \frac{1}{M^2}\left|\sum_{j=1}^{k+1} \omega^{cs_j} e^{2\pi i h s_j/N} D_j(h)\right|^2, \quad D_j(h) := \sum_{m \in B_j} e^{2\pi i m h/M}, \tag{5}$$

and $|D_j(h)| = |\sin(\pi h L_j/M)/\sin(\pi h/M)|$, where $L_j = m_j - m_{j-1}$.

Proof. Substitute $y = cM + h$ into Eq. (4). Then $e^{2\pi i m y/M} = e^{2\pi i m h/M}$ because $mc \in \mathbb{Z}$, and $e^{2\pi i y s_j/N} = \omega^{cs_j} e^{2\pi i h s_j/N}$. Divide by $P_0(cM) = 1/r$ and use $r^2/N^2 = 1/M^2$. □

For a radius $L \geq 0$ define the *neighbourhood ratio* and its worst case:

$$S_c(L) := \sum_{|h| \leq L} R_c(h) = \frac{\Pr[Y \text{ within } L \text{ of } cM]}{\Pr_{\text{ideal}}[Y = cM]}, \quad S^*(L) := \min_{0 \leq c < r} S_c(L). \tag{6}$$

The 2019 choice is $L = k$. The quantity $S_k = \sum_{y \in G_k} P_k(y)$ in the research prompt equals $(1/r)\sum_c S_c(k) \geq S^*(k)$ when $k < M/2$. Except for the explicitly averaged result below, the subsequent neighbourhood bounds are per c and therefore also bound S_k.

4.1 General k literal glitches: a total-mass bound

Answer in brief. After any k literal glitches, the probability of landing in the *set* of ideal outcomes is at least $1/(2k+1)$. If k is small compared with N, a second bound is close to $1/(k+1)$. Both are exact finite-N guarantees. They require neither the block-shift idealization nor a continuum limit.

The key is to count entries by two labels: their **output symbol** u and their **input residue** $b = x \bmod r$, where x is the entry's position. Call each pair (u, b) a *box*. Averaging over all ideal Fourier outcomes

turns interfering amplitudes into a sum of squared box counts. Glitches limit the number of occupied boxes, and spreading N entries among only a few boxes forces that sum of squares to be large.

Theorem 3 (General literal glitches, total mass). new in this note proof below Lean checked
Let N, r be positive integers with $r \mid N$, and start from the infinite periodic sequence with symbols $w_0, ..., w_{r-1}$, which need not be distinct. Perform A single-symbol insertions and D single-symbol deletions, $k = A + D \geq 0$ edits in all, in any order, at arbitrary positions, and with arbitrary inserted symbols. Keep the first N entries of the resulting sequence, obtaining f_k. Let I be the number of surviving inserted entries in this prefix, so $0 \leq I \leq \min(A, N)$.

For the measurement distribution P_{f_k}, define

$$G := \{cN/r : 0 \leq c < r\}, \quad G_k := \left\{ y \in \mathbb{Z}_N : \min_{g \in G} d_N(y, g) \leq k \right\}, \tag{7}$$

where $d_N(y, g) := \min_{m \in \mathbb{Z}} |y - g - mN|$ is circular distance modulo N. Then

$$\Pr[Y \in G_k] \geq \Pr[Y \in G] \geq \frac{r}{(k+1)r + I} \geq \frac{r}{(k+1)r + k} \geq \frac{1}{2k+1}. \tag{8}$$

Independently,

$$\Pr[Y \in G] \geq \frac{(N - I)^2}{(k+1)N^2} = \frac{(1 - I/N)^2}{k+1}. \tag{9}$$

The larger of these two lower bounds may be used. There is no assumption that $k < r$, $k < N$, or $r < \sqrt{N}$. Overlap between neighbourhoods in G_k causes no problem: the proof already bounds the smaller set G. If moreover $r \geq 2$ and $k \geq 1$, then $\Pr[Y \in G_k] \geq 133/358$ by Theorem 5, whose proof does not use $r \nmid N$. No k-independent bound holds on G itself (see the remark below).

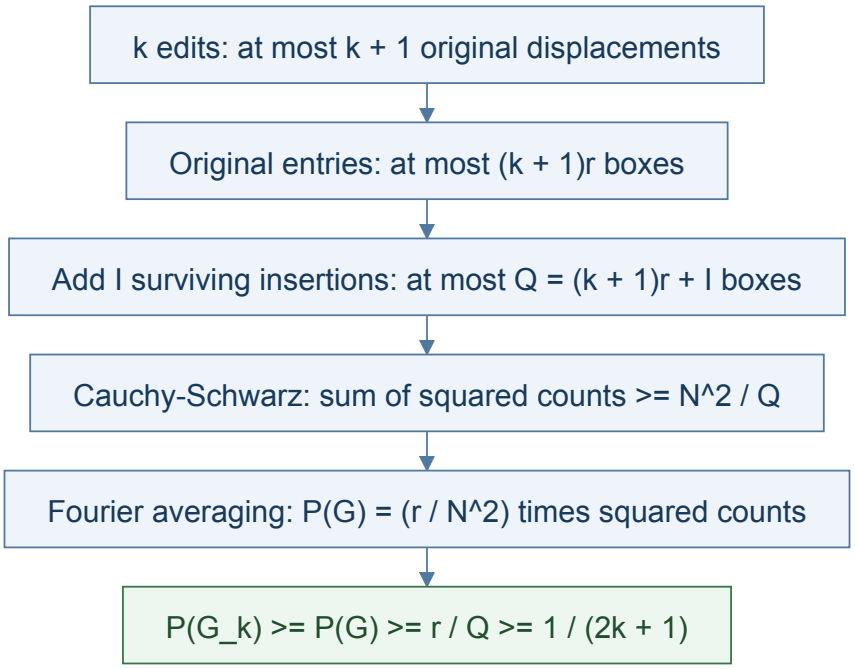


Figure 2: Proof map. A box means one pair (output symbol, input residue modulo r). The count argument controls the *total* probability on G, not its division among the individual good outcomes. Mermaid source: `report_figures/general_k_proof.md`.

Proof. **Step 1: Express the Fourier probability using box counts.**

For any function $f : \mathbb{Z}_N \to W$, let

$$n_{ub} := |\{x \in \{0, ..., N-1\} : f(x) = u, x \equiv b \pmod r\}|. \tag{10}$$

These counts are nonnegative and $\sum_{u,b} n_{ub} = N$. At an ideal outcome $y = cN/r$, the phase depends on x only through its residue b:

$$e^{2\pi i x y/N} = e^{2\pi i c x/r} = e^{2\pi i c b/r}. \tag{11}$$

Thus Eq. (2) becomes

$$P_f(cN/r) = \frac{1}{N^2} \sum_u \left| \sum_{b=0}^{r-1} n_{ub} e^{2\pi i c b/r} \right|^2. \tag{12}$$

Sum over c. The identity

$$\sum_{c=0}^{r-1} e^{2\pi i c(b-b')/r} = \begin{cases} r \text{ if } b = b' \\ 0 \text{ if } b \neq b' \end{cases} \tag{13}$$

removes every cross-term between different residues. Therefore

$$\begin{aligned} \Pr[Y \in G] &= \frac{1}{N^2} \sum_u \sum_{b=0}^{r-1} \sum_{b'=0}^{r-1} n_{ub} n_{ub'} \sum_{c=0}^{r-1} e^{2\pi i c(b-b')/r} \\ &= \frac{r}{N^2} \sum_u \sum_{b=0}^{r-1} n_{ub}^2. \end{aligned} \tag{14}$$

In words: the Fourier calculation has become an ordinary counting problem. Two entries in the same box always reinforce one another in this sum; different boxes do not interfere after averaging over c.

Step 2: Bound the number of occupied boxes.

Give every entry a *tag* recording where it came from. An original entry's tag is its original index $t \geq 0$; the j-th insertion gets the tag $-j$. Write $\tau(x)$ for the tag at position x of the edited sequence. Its symbol is

$$f_k(x) = \begin{cases} w_{\tau(x) \bmod r} & \text{if } \tau(x) \geq 0 \\ v(\tau(x)) & \text{if } \tau(x) < 0 \end{cases} \tag{15}$$

where v assigns the arbitrary inserted symbols. Insertions and deletions preserve the order of all surviving original entries.

Consider an original entry that survives at position x, so that $t = \tau(x) \geq 0$. Among the original t entries preceding it, let $d(t)$ have been deleted. Let $i(t)$ surviving inserted entries precede it in the final sequence. Counting the entries before it gives

$$x = t - d(t) + i(t), \quad s := x - t = i(t) - d(t). \tag{16}$$

Since $0 \leq i(t) \leq I \leq A$ and $0 \leq d(t) \leq D$,

$$-D \leq s \leq A. \tag{17}$$

This interval contains only $A + D + 1 = k + 1$ integers. Hence there are at most $k + 1$ possible displacements, regardless of the order or locations of the edits.

An original entry with symbol w_a has $t \equiv a \pmod{r}$. Its final residue is $x \equiv a + s \pmod{r}$. For each of the r original symbols, at most $k + 1$ residues are therefore possible. So retained original entries occupy at most $(k + 1)r$ boxes. Each of the I surviving inserted entries can add at most one further box. In total,

$$q := |\{(u, b) : n_{ub} > 0\}| \le (k + 1)r + I =: Q. \tag{18}$$

This reasoning includes adjacent edits, several edits inside one period, an insertion subsequently deleted, and inserted symbols equal to original symbols. Reusing an existing box can only reduce the count. Restricting to the first N entries also cannot add boxes.

Step 3: Few boxes force a large sum of squares.

Cauchy–Schwarz, applied just to the q nonempty boxes, gives

$$N^2 = \left(\sum_{u,b} n_{ub} \right)^2 \le q \sum_{u,b} n_{ub}^2 \le Q \sum_{u,b} n_{ub}^2. \tag{19}$$

Equivalently, the sum of squares is smallest when entries are spread as evenly as possible. Substitute into Eq. (14):

$$\Pr[Y \in G] \ge \frac{r}{N^2} \cdot \frac{N^2}{Q} = \frac{r}{(k + 1)r + I}. \tag{20}$$

Since $I \le k$ and $r \ge 1$,

$$\frac{r}{(k + 1)r + I} = \frac{1}{k + 1 + I/r} \ge \frac{1}{2k + 1}. \tag{21}$$

Finally, $G \subseteq G_k$, proving Eq. (8). The same conclusion holds for ordinary, non-wrapped distance-k neighbourhoods, since they too contain G.

Step 4: A second bound, ignoring inserted entries.

Let m_{ub} count only retained original entries in each box. Then $0 \le m_{ub} \le n_{ub}$, their sum is $N - I$, and they occupy at most $(k + 1)r$ boxes. A second application of Cauchy–Schwarz yields

$$\sum_{u,b} n_{ub}^2 \ge \sum_{u,b} m_{ub}^2 \ge \frac{(N - I)^2}{(k + 1)r}. \tag{22}$$

This is also valid when $I = N$, when its right-hand side is zero. Multiplying by r/N^2 in Eq. (14) proves Eq. (9). □

Worked example: three mixed glitches. Start with `abcabcabc…`, insert `d` just before original index 3, insert `e` just before original index 6, and delete original index 9. Keep $N = 15$ entries. Here $r = 3$, $k = 3$, $A = 2$, $D = 1$, and $I = 2$; all positions and indices start at zero. In the edited sequence the three edits happen at positions 3, 7, and 11, and the inserted entries get the tags -1 and -2. Figure 3 tracks the surviving originals and builds the count table.

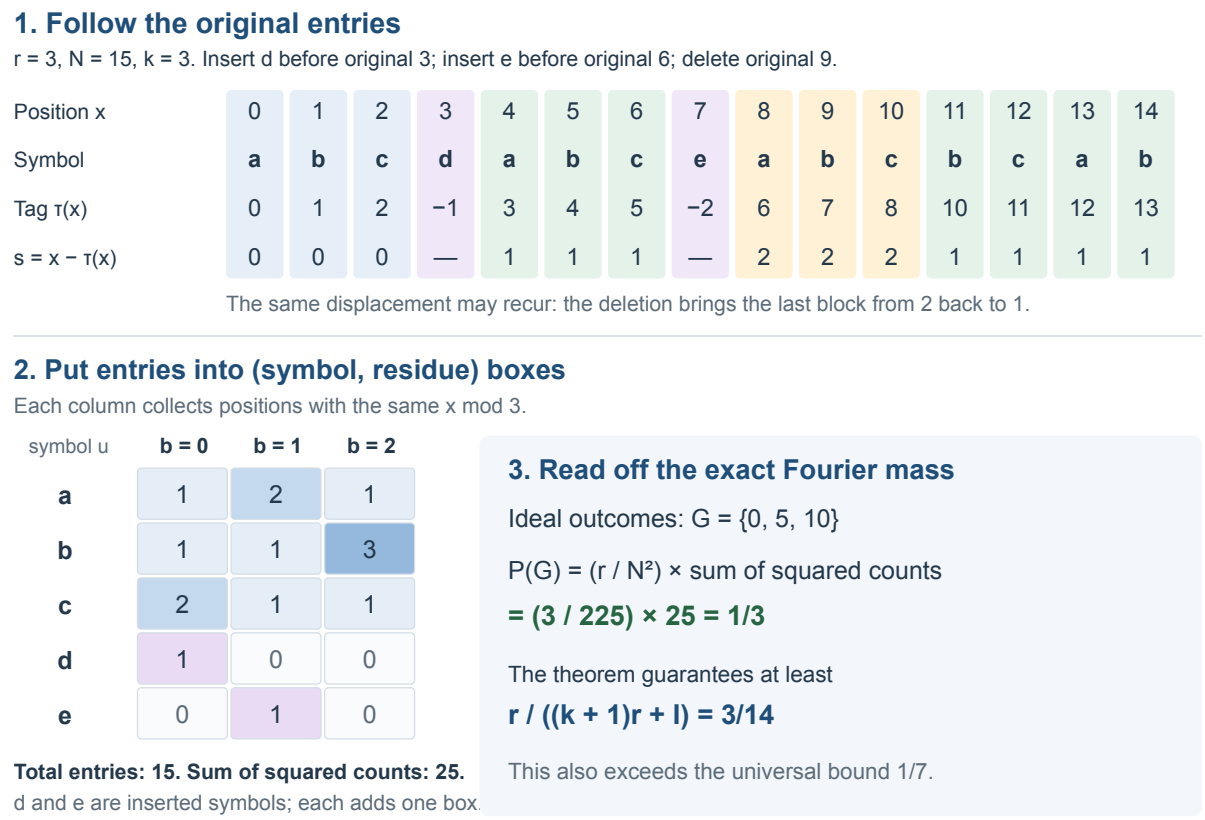


Figure 3: A literal three-edit example, not a block-shift approximation. Inserted entries have negative tags and no displacement. The deletion removes original index 9 and changes the later displacement s from 2 back to 1. The table counts each final entry exactly once.

The table has $\sum n_{ub} = 15$ and $\sum n_{ub}^2 = 25$, so the identity gives

$$\Pr[Y \in \{0, 5, 10\}] = \frac{3}{15^2} \cdot 25 = \frac{1}{3} \geq \frac{3}{(3+1)3+2} = \frac{3}{14} \geq \frac{1}{7}. \tag{23}$$

The middle quantity is the theorem's box-count bound; the last is its simpler universal bound. Repeated displacement values do not need new boxes, so an actual configuration can do better.

Useful special cases.

- **No edits:** $k = I = 0$, and the bound is 1, as it should be.
- **Deletions only (or no inserted entry survives in the prefix):** $I = 0$ gives $\Pr[Y \in G] \geq 1/(k+1)$.
- **Few edits compared with the sample length:** if $k \leq N$, then Eq. (9) gives $\Pr[Y \in G] \geq (1 - k/N)^2/(k+1)$. In particular, for $k = o(N)$ the guarantee is $(1 - o(1))/(k+1)$. The condition $k \leq N$ is needed for this substitution, but not for the theorem itself.

- **Two glitches:** setting $k = 2$ recovers the earlier result $\Pr[Y \in G_2] \geq \Pr[Y \in G] \geq r/(3r + I) \geq r/(3r + 2) \geq 1/5$, together with $\Pr[Y \in G] \geq (N - I)^2/(3N^2)$.

Proposition 4 (General glitches in normalized Model I). new in this note proof below Lean checked For any normalized Model I configuration (in particular, any configuration with $k < r$),

$$\Pr_{\text{Model I}}[Y \in G] \geq \frac{1}{k+1}. \tag{24}$$

This bound is attained at the exact set G when the $k + 1$ blocks have equal lengths and their shifts are distinct modulo r.

Proof. Let $\ell_j = (m_j - m_{j-1})/M$ and group together lengths with the same shift residue:

$$p_b := \sum_{j : s_j \equiv b \pmod r} \ell_j. \tag{25}$$

The p_b are nonnegative, sum to 1, and at most $k + 1$ are nonzero. At $h = 0$, Eq. (5) and root-of-unity orthogonality give

$$\Pr_{\text{Model I}}[Y \in G] = \frac{1}{r} \sum_{c=0}^{r-1} \left| \sum_j \ell_j e^{2\pi i c s_j / r} \right|^2 = \sum_{b=0}^{r-1} p_b^2 \geq \frac{1}{k+1}. \tag{26}$$

For equality, take all insertions, $r > k$, and M divisible by $k + 1$, with $\ell_j = 1/(k + 1)$. The shifts $0, 1, ..., k$ are distinct modulo r, and the $k + 1$ nonzero grouped lengths are all $1/(k + 1)$. □

Remark. **What is proved, and what remains open.** The two probabilities below are different:

- **Total mass:** Theorem 3 guarantees probability $\Omega(1/(k + 1))$ on the union of good outcomes, even before adding their neighbours. Thus the weaker total-mass target in the research prompt is proved for every k, including literal mixed-sign edits.
- **Per-outcome mass:** Conjecture 7 asks for a constant lower bound on $S_\alpha^{\text{cont}(k)}$ for *every* α. Our proof sums over c to cancel cross-terms; it does not bound the summands separately.

Even exact ideal outcomes can have very different probabilities. For example, in Model I with one insertion, two equal blocks, and even $r > 1$, the amplitude at $c = r/2$ is $(1/2) - (1/2) = 0$, although the total mass on G is $1/2$. This example concerns radius zero, not a failure of the radius-k conjecture.

The equality example in Proposition 4 shows that no k-independent positive lower bound is possible on the *exact* set G in Model I. The literal model behaves the same way. Take distinct symbols, $k < r$, and k deletions that split the prefix into $k + 1$ runs of equal length L, a multiple of r. Each symbol then fills $k + 1$ boxes with L/r entries each, so Eq. (14) gives $\Pr[Y \in G] = 1/(k + 1)$ exactly. The larger set G_k does have a constant bound: $133/358$ by Theorem 5, for $r \geq 2$ and $k \geq 1$. Finally, total mass includes uninformative outcomes such as $c = 0$: neither theorem supplies the mass near coprime c, so period recovery remains a separate problem (Section 9).

Machine-checked companion. The proofs for this subsection are in `formal/`, using Lean 4 and mathlib v4.19.0. The formalization starts with actual insertion/deletion operations on an infinite tagged sequence, proves the displacement and box-count bounds, proves character orthogonality, and connects

the counts to the complex-exponential Fourier measurement formula. It does not assume the key counting or Fourier identities as hypotheses.

The checked results include both bounds in Theorem 3, their neighbourhood and finite-size corollaries, the constant 133/358 on G_k (`general_k_dividing_windows`), and the Model I bound and sharpness in Proposition 4. The figure's actual edit history, count table, and mass 1/3 are also checked, not just its final fraction. There are no `sorry` placeholders or added axioms; the axiom audit reports only Lean's standard logical foundations.

The formal alphabet is finite, which suffices because a finite period and finitely many edits use only finitely many symbols. The Fourier measurement formula is taken as the definition; a quantum circuit implementation is not verified. Lean also covers Section 4.2 below; it does not certify the other continuum results, the transfer sketch, or the open per-outcome and recovery claims. The Lean names follow this note's symbols. See `formal/README.md` for the theorem-by-theorem map and a table matching each symbol to its Lean name, and Appendix A for commands.

4.2 Nondivisible periods: a constant neighbourhood bound

The exact box-count identity requires $r \mid N$; substituting rounded peak locations into that identity is not valid. Nevertheless, the *total neighbourhood mass* has a stronger bound that does not require divisibility. The following argument uses the actual literal edited prefix, not the legacy nondivisible Model I formula with $M = \lceil N/r \rceil$.

Theorem 5 (Literal glitches without a divisibility assumption). new in this note proof below Lean checked Let $N \geq 1$ and $r \geq 2$. Start from an infinite r-periodic sequence, perform A insertions and D deletions at arbitrary positions, $k = A + D \geq 1$ edits in all, and keep the first N entries, obtaining f_k. Inserted symbols are arbitrary. Let g_c be any nearest integer to cN/r, read modulo N, and put

$$G_k := \left\{ y \in \mathbb{Z}_N : \min_{0 \leq c < r} d_N(y, g_c) \leq k \right\}. \tag{27}$$

For the Fourier measurement distribution Eq. (2),

$$\Pr[Y \in G_k] \geq \frac{133}{358} > \frac{1}{3} \geq \frac{1}{2k+1}. \tag{28}$$

This holds whether or not $r \mid N$, for all edit placements and mixed signs. There is no additional assumption on k/N, kr/N, or r/N, and overlapping windows are counted only once. The constant 133/358 is not claimed to be sharp.

Proof. **Step 1: Relate Fourier moments to equal-symbol pairs.**

For any integer $h \geq 1$, define the circular matching fraction

$$C_h := \frac{1}{N} \sum_{x=0}^{N-1} \mathbb{1}[f_k(x) = f_k((x+h) \bmod N)]. \tag{29}$$

Expanding Eq. (2) and summing the N roots of unity gives

$$\sum_{y=0}^{N-1} P_{f_k}(y) \cos(2\pi h y/N) = C_h. \tag{30}$$

Use the tags τ from Step 2 of Section 4.1. Call x a *good start* for the lag h if positions x and $x+h$ both hold original entries whose tags differ by exactly h, that is, $\tau(x) \geq 0$ and $\tau(x+h) = \tau(x) + h$. When $r \mid h$ the two tags are congruent modulo r, so a good start has $f_k(x) = f_k(x+h)$.

For every length L, at most $kh + A$ of the starts $x < L$ are not good. This follows by induction on the edit history. In the unedited sequence every start is good. An insertion destroys at most h good pairs, those that straddle it, and adds one start. A deletion destroys at most $h+1$ good pairs, those that contain or straddle the deleted entry, and removes one start. So each insertion adds at most $h+1$ starts that are not good, each deletion at most h, and $A(h+1) + Dh = kh + A$.

Take $L = N - h$, so that no pair wraps around, and count the h wrapped starts as failures. For every multiple h of r this gives

$$C_h \geq 1 - \frac{(k+1)h + A}{N}. \tag{31}$$

For $h \geq N$ the right-hand side is nonpositive, so the inequality is automatic.

Step 2: Separate the rounded good windows from their complement.

If $N < (2k+1)r$, the rounded windows cover every integer outcome. Indeed, an integer is within $N/(2r)$ of some real peak and within $N/(2r) + 1/2 < k+1$ of its rounded peak, hence within k. Thus this case is immediate. Otherwise $N \geq (2k+1)r$; put

$$t := \frac{2N}{r(2k+1)} \geq 2, \quad p := \Pr[Y \in G_k], \quad \theta_y := \min_{m \in \mathbb{Z}} |ry/N - m| \in \left[0, \frac{1}{2}\right]. \tag{32}$$

If y is outside G_k, then its integer distance from every g_c is at least $k+1$. Since $d_N(g_c, cN/r) \leq 1/2$, its distance from every real peak is at least $k + 1/2$. Consequently $\theta_y \geq 1/t$.

For every integer $\ell \geq 1$, Eq. (31), $r \geq 2$, and $A \leq k$ imply

$$C_{\ell r} \geq 1 - \frac{2(k+1)\ell + k}{(2k+1)t} \geq 1 - \frac{4\ell + 1}{3t}. \tag{33}$$

The last inequality is equivalent to $(k-1)(2\ell - 1) \geq 0$.

Step 3: Average a nonnegative Fourier kernel.

For an integer $T \geq 2$, the Fejér kernel is

$$F_T(\theta) := \frac{1}{T}\left|\sum_{j=0}^{T-1} e^{2\pi i j\theta}\right|^2 = 1 + 2\sum_{\ell=1}^{T-1}(1 - \ell/T)\cos(2\pi\ell\theta). \tag{34}$$

It satisfies $0 \leq F_T \leq T$. From Eq. (30) and Eq. (33),

$$E_T := \sum_y P_{f_k}(y) F_T(ry/N), \quad \frac{E_T}{T} \geq 1 - \frac{4(T^2-1)}{9tT} - \frac{T-1}{3tT}. \tag{35}$$

Here $\sum_{\ell=1}^{T-1} \ell(1 - \ell/T) = (T^2-1)/6$ and $\sum_{\ell=1}^{T-1}(1 - \ell/T) = (T-1)/2$.

If $F_T/T \leq q < 1$ outside G_k, then

$$\frac{E_T}{T} \le p + (1-p)q, \quad p \ge \frac{E_T/T - q}{1-q}. \tag{36}$$

Step 4: Choose the kernel size, using two elementary estimates.

Short kernel, $2 \le t \le 6$. Take $T = 2$. Then $F_2(\theta)/2 = \cos^2(\pi\theta)$, so we may take $q = \cos^2(\pi/t)$. Eq. (35) gives $E_2/2 \ge 1 - 5/(6t)$, hence

$$p \ge 1 - \frac{5}{6t\sin^2(\pi/t)} \ge \frac{3}{8}. \tag{37}$$

To check the last step, $t\sin^2(\pi/t) \ge 4/3$ throughout this range: for $2 \le t \le 3$ use $\sin^2(\pi/t) \ge 3/4$; for $3 \le t \le 4$ use $\sin^2(\pi/t) \ge 1/2$; for $4 \le t \le 6$, concavity of sine on $[0, \pi/4]$ gives $\sin(\pi/t) \ge 2\sqrt{2}/t$, so $t\sin^2(\pi/t) \ge 8/t \ge 4/3$.

Long kernel, $t > 6$. Take $T = \lceil 3t/4 \rceil$. Using $T \le 3t/4 + 1$ in Eq. (35) gives

$$\frac{E_T}{T} \ge \frac{2}{3} - \frac{7}{9t} \ge \frac{29}{54}. \tag{38}$$

Outside G_k, the geometric-sum formula gives

$$F_T(\theta_y)/T \le \frac{1}{T^2\sin^2(\pi\theta_y)} \le \frac{1}{T^2\sin^2(\pi/t)} \le \frac{64}{243}. \tag{39}$$

Indeed, concavity of sine on $[0, \pi/3]$ gives $\sin(\pi/t) \ge \left(3\sqrt{3}/2\right)/t$, and $T \ge 3t/4$. Now Eq. (36) yields

$$p \ge \frac{29/54 - 64/243}{1 - 64/243} = \frac{133}{358} > \frac{1}{3}. \tag{40}$$

Since $3/8 > 133/358$, both cases prove the stated bound. □

Remark. **Scope.** This proves the original $1/(2k+1)$ *total-mass* guarantee for $r \nmid N$ and $k \ge 1$, without an extra finite-size condition. It does not extend the box-count identity, bound the mass at just the rounded peaks, or bound each peak separately. In particular, it still does not establish period recovery or the informative-outcome conjecture. The $k = 0$ claim with lower bound 1 cannot be carried over to the nearest integers: an unedited nondivisible period already has Fourier leakage.

Machine-checked companion. `formal/Glitchy/Nondividing.lean` proves this theorem end to end for the same literal edit histories and complex-exponential measurement formula. Matching moments, pair counts, rounded-window separation, the squared geometric sum, and both analytic regimes are proved, not supplied as hypotheses. Every rounding-tie choice is allowed, and overlapping windows are counted as a set. The axiom audit uses only Lean's standard logical foundations, with no placeholders or added axioms.

The Lean proof follows Steps 1–4 above with the same symbols: the good-start induction, the coverage case $N < (2k+1)r$, and the short and long kernels. Deterministic numerical checks remain in `glitchy.py check`. See `formal/README.md` for the declaration map.

4.3 Continuum limit

Let $u_j = m_j/M \in [0,1]$, $\ell_j = u_j - u_{j-1}$ and $\alpha = c/r$. Define the *phase function* and its Fourier coefficients

$$\varphi_\alpha(u) = e^{2\pi i \alpha s_j} \quad \text{for } u \in [u_{j-1}, u_j),$$
$$a_h := \int_0^1 \varphi_\alpha(u) e^{2\pi i h u}\,\mathrm{d}u = \sum_{j=1}^{k+1} e^{2\pi i \alpha s_j} e^{i\pi h(u_{j-1}+u_j)} \frac{\sin(\pi h \ell_j)}{\pi h}, \tag{41}$$

where the $h = 0$ term reads ℓ_j. The continuum neighbourhood mass is $S_\alpha^{\mathrm{cont}}(L) = \sum_{|h|\le L} |a_h|^2$.

Lemma 6 (Continuum approximation). new in this note proof below For all c and h, with $\varepsilon_h := \pi|h|(1 + 2k/r)/M$,

$$\left|\sqrt{R_c(h)} - |a_h|\right| \le \varepsilon_h, \quad \left|S_c(L) - S_{c/r}^{\mathrm{cont}}(L)\right| \le \sum_{|h|\le L} (2\varepsilon_h + \varepsilon_h^2) = O(L^2(1 + k/r)/M). \tag{42}$$

Proof. $\sqrt{R_c(h)}$ is the modulus of $A := M^{-1}\sum_j \omega^{cs_j} e^{2\pi i h s_j/N} D_j(h)$. The sum $M^{-1}D_j(h)$ is a left Riemann sum, with step $1/M$, for $\int_{u_{j-1}}^{u_j} e^{2\pi i h u}\,\mathrm{d}u$. The integrand has derivative of size at most $2\pi|h|$, so each cell contributes error at most $\pi|h|/M^2$, and the total over all blocks is at most $\pi|h|/M$. In addition, $\left|e^{2\pi i h s_j/N} - 1\right| \le 2\pi|h|k/N = 2\pi|h|k/(rM)$, and $\sum_j |D_j|/M \le 1$. So $|A - a_h| \le \varepsilon_h$. Finally, $\left||A|^2 - |a_h|^2\right| \le \varepsilon_h(2 + \varepsilon_h)$ because $|a_h| \le 1$. □

4.4 The poster's "gruesome expression"

The poster states the multi-glitch problem with $l = k$ glitches at $0 < x_1 < \ldots < x_l < t$ and $0 < \alpha < 1$. Write $t = 1$, $x_j = u_j$, and take all insertions, so $s_j = j - 1$. Its quantities are the continuum Fourier data of Eq. (41):

$$A_0 = |a_0|^2, \quad A_{hj} = \frac{\sin(\pi h \ell_j)}{\pi h}, \quad S_h = |a_h|^2 + |a_{-h}|^2, \quad S = A_0 + \sum_{h=1}^{l} S_h = S_\alpha^{\mathrm{cont}}(l). \tag{43}$$

Expanding the squares gives the poster's formulas:

$$\begin{aligned} A_0 &= \sum_{j=1}^{l+1} \ell_j^2 + 2 \sum_{1\le j'<j\le l+1} \ell_j \ell_{j'} \cos(2\pi\alpha(j - j')), \\ S_h &= 2\sum_{j=1}^{l+1} A_{hj}^2 + 4 \sum_{1\le j'<j\le l+1} A_{hj}A_{hj'} \cos(2\pi\alpha(j - j')) \cos\big(\pi h\big(u_{j-1} + u_j - u_{j'-1} - u_{j'}\big)\big). \end{aligned} \tag{44}$$

The second line uses $\cos(\theta + \psi) + \cos(\theta - \psi) = 2\cos\theta\cos\psi$. Three transcription slips on the poster are corrected here. The first sum in A_0 runs over $j = 1, \ldots, l + 1$, not $0, \ldots, l$. The last term in the second cosine is $x_{j'}$, not x_j. The prefactor t in A_{hj} is consistent only with the normalization $t = 1$. For mixed signs, replace $j - j'$ by $s_j - s_{j'}$. All-deletion configurations give the same S, because S is even in α.

Conjecture 7 (Poster, 2019). conjecture from 2019 work For every $k \ge 1$, every $0 < u_1 < \ldots < u_k < 1$, all-insertion or all-deletion signs, and every $\alpha \in [0, 1)$, we have $S_\alpha^{\mathrm{cont}}(k) \ge 1/2$. By Lemma 6 this gives $S^*(k) \ge 1/2 - O(k^2(1 + k/r)/M)$ in Model I.

The poster interprets this as "even in the worst case, the cost to the runtime would scale like the number of glitches". The $2k + 1$ candidates around each outcome must be tried, but only $O(1)$ more samples are needed (Section 9). The target $S_k \ge c/(k + 1)$ in the research prompt is weaker than Conjecture 7:

Section 4.1 proves it in normalized Model I with constant 1 and in the literal model with constant $1/2$, since $1/(2k+1) \geq (1/2)/(k+1)$.

5 A Parseval reformulation

Proposition 8 (Energy and jump formula). new in this note proof below Let $\varphi_j := e^{2\pi i \alpha s_j}$ be the value of φ_α on piece j. Set $u_0 := 0$ and define the jumps

$$\Delta_j := \varphi_j - \varphi_{j+1} \quad (1 \leq j \leq k), \qquad \Delta_0 := \varphi_{k+1} - \varphi_1 \quad (\text{wrap-around at } u = 0 \equiv 1). \tag{45}$$

1. $\sum_{h\in\mathbb{Z}} |a_h|^2 = 1$, so $S_\alpha^{\text{cont}}(L) = 1 - \sum_{|h|>L} |a_h|^2$.
2. For $h \neq 0$, $a_h = (2\pi i h)^{-1} \sum_{j=0}^{k} \Delta_j e^{2\pi i h u_j}$.
3. $|\Delta_j| = 2|\sin \pi\alpha|$ for every glitch $j \geq 1$, and $|\Delta_0| = 2|\sin(\pi\alpha s_{k+1})|$.

Consequently

$$1 - S_\alpha^{\text{cont}}(L) = \frac{1}{4\pi^2} \sum_{|h|>L} \frac{1}{h^2} \left| \sum_{j=0}^{k} \Delta_j e^{2\pi i h u_j} \right|^2. \tag{46}$$

Proof. (1) Parseval's identity for $\varphi_\alpha \in L^2[0,1)$ with $|\varphi_\alpha| = 1$. (2) Integrate each piece, $\int_{u_{j-1}}^{u_j} e^{2\pi i h u}\, du = \big(e^{2\pi i h u_j} - e^{2\pi i h u_{j-1}}\big)/(2\pi i h)$, and collect the coefficient of each $e^{2\pi i h u_j}$. Use $e^{2\pi i h \cdot 1} = 1$ to merge the endpoint terms into Δ_0. (3) $\Delta_j = \varphi_j\big(1 - e^{2\pi i \alpha \varepsilon_j}\big)$ with $\varepsilon_j = \pm 1$, and $s_1 = 0$. □

So the multi-glitch problem is purely about step functions. S is the fraction of the energy of a unimodular step function with $k+1$ jumps that lies at frequencies $|h| \leq L$. The k glitch jumps have equal size, and there is one wrap-around jump. All destructive interference is in the exponential sum $\sum_j \Delta_j e^{2\pi i h u_j}$ in Eq. (46).

Corollary 9 (Quadratic radius suffices). new in this note proof below $S_\alpha^{\text{cont}}(L) \geq 1 - 2(k+1)^2/(\pi^2 L)$ for every configuration and every α. In particular, $S_\alpha^{\text{cont}}(L) \geq 1/2$ for $L \geq 4(k+1)^2/\pi^2$.

Proof. In Eq. (46) use $\left|\sum_j \Delta_j e^{2\pi i h u_j}\right| \leq \sum_j |\Delta_j| \leq 2(k+1)$ and $\sum_{|h|>L} h^{-2} \leq 2/L$. □

This bound gives nothing at $L = k$, because $2(k+1)^2/(\pi^2 k) > 1$ for every $k \geq 1$. The crude estimate ignores cancellation between jumps. Exploiting that cancellation, for example with large-sieve-type inequalities for well-separated jump positions combined with merging of clustered jumps, is the obvious route to radius $O(k)$.

6 One glitch

Theorem 10 (One glitch). 2019 claim, proof not on file new in this note proof below Let $k = 1$, with the glitch at $u \in (0,1)$, either sign, and any $\alpha \in [0,1)$. In the continuum limit,

$$S_\alpha^{\mathrm{cont}}(1) = 1 - (1 - \cos 2\pi\alpha) D(u), \quad D(u) := 2u(1-u) - \frac{4}{\pi^2}\sin^2(\pi u) \geq 0. \tag{47}$$

Hence

$$S_\alpha^{\mathrm{cont}}(1) \geq g(u) := 1 - 4u(1-u) + \frac{8}{\pi^2}\sin^2(\pi u), \tag{48}$$

with equality when $\alpha = 1/2$, and $g(u) \geq 8/(8+\pi^2) = 0.44768...$ for all u. The sharp constant is $\min_u g(u) = 0.6400$, attained at $u \approx 0.198$ and its mirror $u \approx 0.802$.

Proof. With $\theta = 2\pi\alpha\varepsilon_1$ we have $\Delta_1 = 1 - e^{i\theta}$ and $\Delta_0 = e^{i\theta} - 1$. So $\left|\Delta_1 e^{2\pi i h u} + \Delta_0\right|^2 = \left|1 - e^{i\theta}\right|^2 \left|e^{2\pi i h u} - 1\right|^2 = 8(1-\cos\theta)\sin^2(\pi h u)$. By Eq. (46), $1 - S = 2(1 - \cos\theta)\pi^{-2}\sum_{|h|\geq 2}\sin^2(\pi h u)/h^2$. The classical identity $\sum_{h\geq 1}\sin^2(\pi h u)/h^2 = \pi^2 u(1-u)/2$ for $u \in [0,1]$ gives $\sum_{|h|\geq 2}\sin^2(\pi h u)/h^2 = \pi^2 u(1-u) - 2\sin^2(\pi u)$. Since $\cos\theta = \cos 2\pi\alpha$, Eq. (47) follows.

For $D \geq 0$, use the symmetry $u \leftrightarrow 1-u$ and take $u \leq 1/2$. If $u \leq 1/3$, then $\sin(\pi u) \leq \pi u \leq \pi\sqrt{u(1-u)/2}$. If $1/3 \leq u \leq 1/2$, then $\sin(\pi u) \leq 1 \leq \pi/3 \leq \pi\sqrt{u(1-u)/2}$. Hence the worst α is $1/2$, which gives g.

Jordan's inequality $\sin(\pi u) \geq 2u$ on $[0, 1/2]$ gives $g(u) \geq 1 - 4u + (4 + 32/\pi^2)u^2$. This quadratic is minimized at $u = 2/(4 + 32/\pi^2)$, where it equals $1 - 4/(4 + 32/\pi^2) = 8/(8+\pi^2)$. □

Remark. **Relation to the poster.** The poster says the probability is "greater than 0.447" times the no-glitch probability. The constant $8/(8+\pi^2) = 0.44769$ agrees in every printed digit and comes from the natural argument above: reduce to $\alpha = 1/2$, then apply Jordan's inequality. We therefore believe this was the 2019 argument, but the original proof is not in the repository. Jordan's inequality is the only lossy step. The sharp continuum value is 0.6400. An exhaustive search over the glitch block in Model I gives 0.6403 for $(N, r) = (3000, 50)$ and 0.6400 for $(20000, 20)$. The worst case $\alpha = 1/2$ occurs only for even r and $c = r/2$. Odd r does slightly better. The theorem is stated in the continuum limit. In Model I, Lemma 6 gives $S^*(1) \geq 0.4476 - O((1 + 1/r)/M)$. Since the bound holds for each c separately, the probability of landing within 1 of a good outcome cM with $\gcd(c, r) = 1$ is at least $0.447 \cdot \varphi(r)/r$.

7 Numerical evidence

7.1 The simulator

Command	What it does
`glitchy.py plot`	Plots, for one placement (given or random), the probability near every good outcome relative to ideal: the neighbourhood sum, the exact good outcome, and a heatmap over the offsets $\|h\| \le k$.
`glitchy.py worst`	Searches every placement of k glitches, or a random sample, for any k and sign pattern. Lists the placements with the smallest $\min_c S_c(k)$.
`glitchy.py check`	Reproduces the numerical results and checks the new total-mass bounds in a few seconds.
`Experiment_plots/Log_Plot{2,3,4,5}.txt`	2019 logs: placement (`ListK`), shifts, and the 20 worst placements. The plot images were too large to commit.

Table 1: Code in the repository. `glitchy.py` (2026) replaces the 2019 scripts `Plot.py`, `Find_Worst_Case.py` and `Test.py`, which remain in git history (commit `a9e171b`). It computes the same distribution: on test configurations, including $r \nmid N$, it agrees with the 2019 formula to 10^{-16}. The `Test.py` experiment (adding one extra glitch) had no logged output and was not ported. Good outcomes are the integers nearest cN/r, the same set as the 2019 rule $(yr \bmod N) \le \lfloor r/2 \rfloor$ or $\ge N - \lfloor r/2 \rfloor$. They equal cM when $r \mid N$.

7.2 Worst cases

Log	N	r	M	k	signs	placements	$S^*(k)$	at c/r	continuum
—	3000	50	60	1	ins.	all 59	0.6403	0.50	0.6400
Plot2	3000	50	60	2	ins.	all 1711	0.7723	0.70	0.7720
Plot3	3000	100	30	3	del.	all 3654	0.6136	0.50	0.6119
Plot4	1500	50	30	4	ins.	all 23751[†]	0.6888	0.40	0.6858
Plot5	3000	50	60	5	del.	5063 (grid)	not logged	—	—

Table 2: Worst neighbourhood ratio $S^*(k) = r \cdot \min_c \sum_{|h| \le k} P_k(cM + h)$ in Model I, over all placements with the given signs, and the good outcome c/r where it is attained. The worst placements (block indices $m_1, ..., m_k$) are $(6, 54)$ for Plot2, $(3, 6, 27)$ for Plot3 and $(2, 4, 6, 8)$ for Plot4, whose mirror image $(22, 24, 26, 28)$ ties. The last column evaluates the same configuration in the continuum limit. [†]The 2019 log searched only 1820 grid placements and found 0.7229 at $(3, 24, 25, 28)$. The full search, the $k = 1$ row and the c/r column were computed for this note. Plot3 and Plot4 have $r > \sqrt{N}$, which does not matter when $r \mid N$ but violates the standing assumption. For Plot2 and Plot3 the full search finds the same minima and minimizers as the 2019 logs.

All worst cases stay well above 1/2. The worst good outcome is $c = r/2$ ($\alpha = 1/2$) for $k = 1, 3$, but $\alpha = 0.7$ (equivalently 0.3, since S is even in α) for $k = 2$ and $\alpha = 0.4$ for $k = 4$. So $\alpha = 1/2$ is not always the worst case. The $k = 4$ minimizer $(2, 4, 6, 8)$ packs the glitches at one end, like the family in Section 8.1. The values are not monotone in k: $0.640, 0.772, 0.614, 0.689$ for $k = 1, 2, 3, 4$, though at different N and r. Any induction on k has to account for the radius growing with k.

7.3 Sign patterns

At $\alpha = 1/2$ the phase function is $\varphi_j = e^{i\pi s_j} = (-1)^{j-1}$ whatever the signs, because a shift of ± 1 changes the phase by $e^{\pm i\pi} = -1$. So in the continuum limit $S_{1/2}$ does not depend on the sign pattern; signs matter only for $\alpha \neq 1/2$. Searching all placements with $N = 3000$, $r = 50$ numerical evidence only gives:

signs	$++$	$+-$	$+++$	$++-$	$+-+$	$+--$
$S^*(k)$	0.7723	0.7830	0.6083	0.6083	0.6083	0.6083
at c/r	0.70	0.50	0.50	0.50	0.50	0.50

Flipping every sign gives the same values, because S is even in α. In these runs mixed signs are never worse than equal signs. For $k = 3$ the worst case sits at $\alpha = 1/2$, where signs cannot matter.

7.4 Figures from the poster

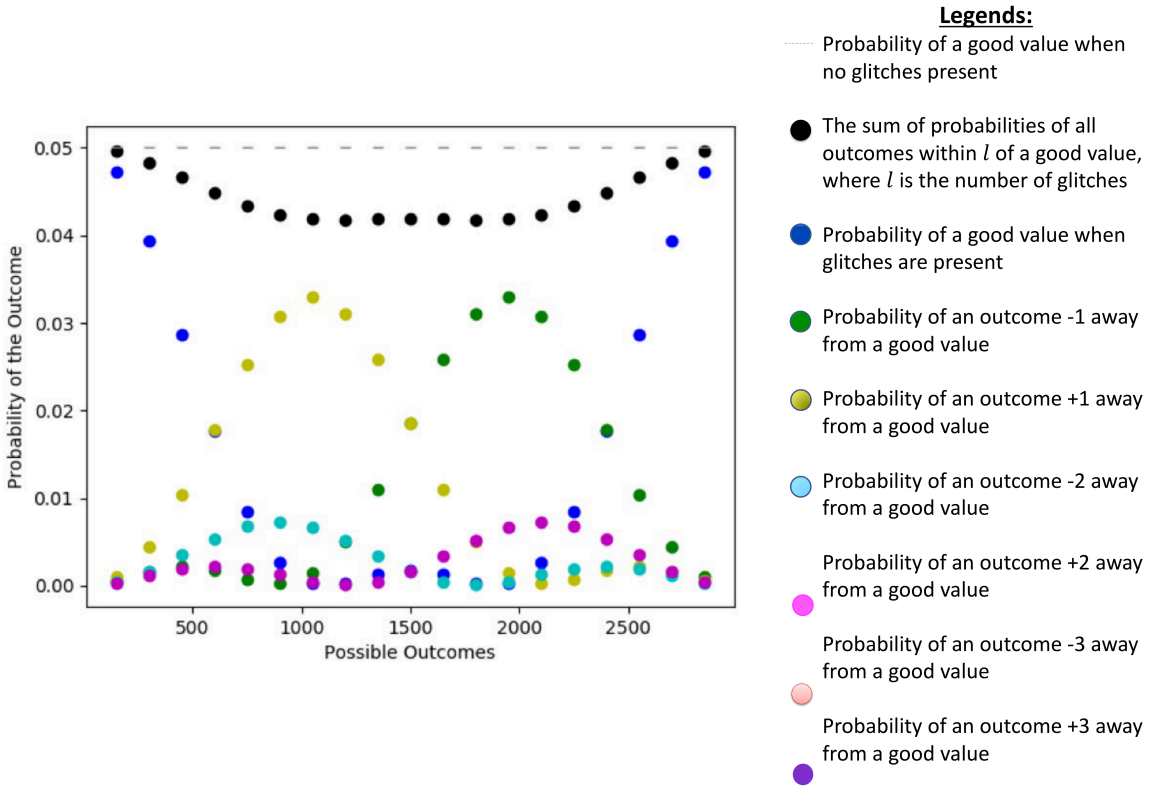


Figure 4: Poster figure 3: $N = 3000$, $r = 20$, $k = 2$. Each column is one good outcome cM. Blue is $P_k(cM)$. Green and yellow are $P_k(cM \mp 1)$, and cyan and magenta are $P_k(cM \mp 2)$. Black is the sum over $|h| \leq 2$. The dotted line is the ideal $1/r = 0.05$. Where the good outcome loses probability, its neighbours gain it.

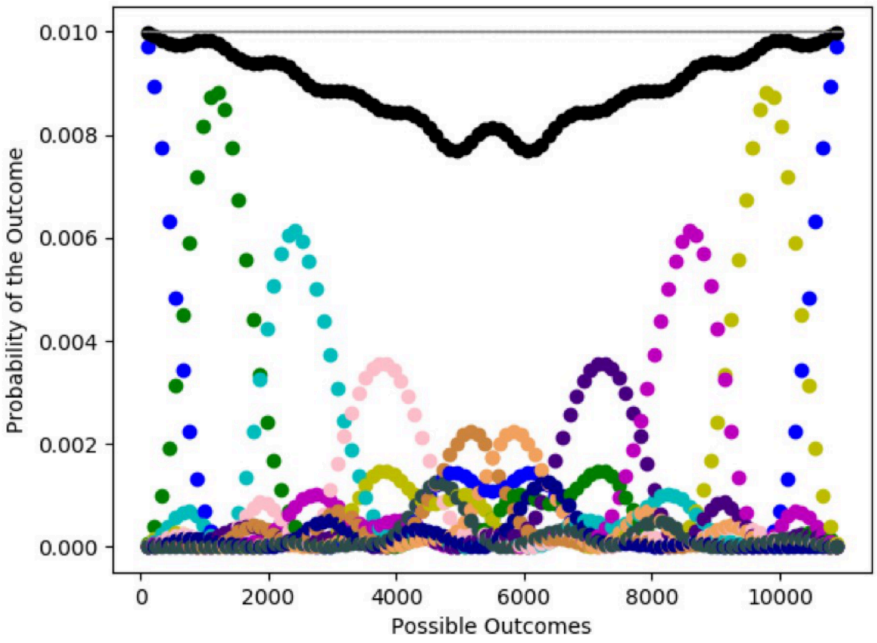


Figure 5: Poster figure 4: $N = 11000$, $r = 100$, $k = 10$, one random placement. The black neighbourhood sum ($|h| \le 10$) stays above about 0.0077, which is 0.77 of the ideal 0.01. The minima are near, but not at, $c = r/2$. The placement was not logged.

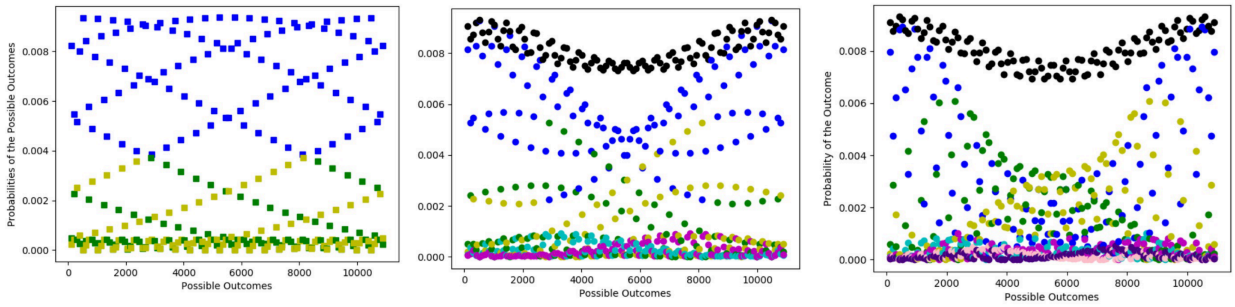


Figure 6: Poster figure 5: $N = 11000$, $r = 107$ ($r \nmid N$) with no glitch (left), 2 glitches (middle) and 3 glitches (right). Without glitches the nearest-integer good outcomes already carry between roughly $4/\pi^2$ and 1 times $1/r$. No analysis exists for this case.

8 Further observations made while compiling this note

8.1 A packed family approaching 1/2

Place all k glitches, with equal signs, equally spaced in $[0, \delta)$, and take $\alpha = 1/2$ (even r, $c = r/2$). Minimizing $S_{1/2}^{\text{cont}}(k)$ over a grid of δ values numerical evidence only gives:

k	2	4	6	10	20	50	100	200
$\min_\delta S$	0.782	0.696	0.653	0.611	0.571	0.542	0.531	0.521
δ	0.245	0.330	0.375	0.415	0.455	0.480	0.490	0.490

Heuristically, on $[0, \delta)$ the phase alternates ± 1 every δ/k. This is a square wave of frequency $k/(2\delta)$, which exceeds k when $\delta < 1/2$. Its energy, about δ, therefore lies outside the window $|h| \le k$, while

the constant part on $[\delta, 1)$ keeps about $1-\delta$. Letting $\delta \uparrow 1/2$ as $k \to \infty$ suggests $\inf S^{\mathrm{cont}}(k) \to 1/2$. **Consequence:** if Conjecture 7 is true, its constant is asymptotically sharp, and a proof must be tight on this family. Neither the limit nor the existence of configurations below $1/2$ has been settled.

8.2 The radius must grow linearly in k

Proposition 11 (Radius lower bound). new in this note proof below Let k be odd, the glitches equally spaced ($u_j = j/(k+1)$), and the signs equal.

1. If $\alpha = 1/2$, then $a_h = 0$ unless h is an odd multiple of $(k+1)/2$. Hence $S^{\mathrm{cont}}_{1/2}(L) = 0$ for $L < (k+1)/2$, and $S^{\mathrm{cont}}_{1/2}(L) = 8/\pi^2$ for $(k+1)/2 \le L < 3(k+1)/2$.
2. If r is odd and $c = (r-1)/2$, which is informative because $\gcd(c, r) = 1$, then $S^{\mathrm{cont}}_{c/r}(L) \le (2L+1)(\pi k/r)^2$ for $L < (k+1)/2$.

Proof. (1) The $k+1$ pieces alternate ± 1, and since $k+1$ is even this holds across the wrap-around too. So $\varphi(u+\tau) = -\varphi(u)$ with $\tau = 1/(k+1)$, and therefore $a_h = -e^{2\pi i h\tau} a_h$. Hence $a_h = 0$ unless $e^{2\pi i h\tau} = -1$. The energy at the fundamental frequency of a ± 1 square wave is $8/\pi^2$. (2) $|a_h(\alpha) - a_h(1/2)| \le \left\|\varphi_\alpha - \varphi_{1/2}\right\|_1 \le \max_j \left|e^{2\pi i(\alpha-1/2)s_j} - 1\right| \le 2\pi|\alpha - 1/2|k = \pi k/r$. Then apply (1). □

So any bound that is uniform in c, or that holds near the informative $c \approx r/2$, needs radius $L \ge (k+1)/2$ (when $r \gg k^{3/2}$). The same configuration also suggests a trade-off. For a general α its energy concentrates near $h \equiv -\alpha(k+1) \pmod{k+1}$, so a radius L captures roughly a fraction $2L/(k+1)$ of the values of c. The total cost, (samples) × (candidates per sample), is then about k for every choice of L. This is heuristic.

8.3 From the literal model to Model I

Proposition 12 (Transfer). new in this note proof sketch only Let f be a literal k-glitch function with $k < r$ (Section 3, remark on idealizations), and round each glitch to the next period boundary to obtain a Model I configuration. Then for all c and L,

$$\left|S^{\mathrm{literal}}_c(L) - S^{\mathrm{Model\ I}}_c(L)\right| = O(Lk(1 + k/M)/M). \tag{49}$$

Together with Lemma 6, a continuum bound $S^{\mathrm{cont}}(k) \ge s_0$ gives $S^{\mathrm{literal}}(k) \ge s_0 - O(k^2(1 + k/r)/M)$. Since $M > \sqrt{N}$, this is meaningful for $k = o(N^{1/4})$.

Sketch. For each class a the two position sets differ in δ_a points (moved, added or removed), with $\delta_a = O(k)$ and $\sum_a \delta_a = O(kr)$. A glitch inside a period moves at most $r-1$ points, one per class. Inserted symbols and truncation contribute $O(k)$ more. New symbols form singleton classes, which add at most k/N^2 to each $P(y)$. Since each class amplitude changes by at most $2\delta_a$ and has size at most $M + \delta_a$, $|P(y) - P'(y)| \le N^{-2} \sum_a 2\delta_a(2M + 2\delta_a) = O(k(M+k)r/N^2)$. Multiplying by $r(2L+1)$ gives the claim. The bookkeeping of δ_a, in particular several glitches in one period and mixed signs, still has to be written out. □

9 From neighbourhood mass to period recovery

The poster's hypothesis is that the period can be learned by examining every point within a certain range of the measured outcome. Made explicit, the procedure is proposal, not proved:

1. Run the period-finding circuit on f_k and measure y.

2. For each y' with $|y' - y| \le L$, expand y'/N in continued fractions and collect the convergent denominators $q \le \sqrt{N}$ as candidates.
3. For each candidate, sample random x and test $f_k(x) = f_k(x+q)$. Output the smallest q whose pass rate exceeds a threshold $1 - \tau$.

Why it should work ($r \mid N$, Model I):

- If y lands within L of cM with $\gcd(c, r) = 1$, then $y' = cM$ is in the window and $y'/N = c/r$ exactly, so $q = r$ is a candidate.
- For $q = r$, the test fails only when a glitch lies in $(x, x + r]$. This part is proved for literal edits and any r, divisible or not. The pair count behind Eq. (31) shows that at most $kr + A$ of the $N - r$ pairs $(x, x + r)$ break, so $q = r$ passes whenever $\tau > (kr + A)/(N - r)$.
- For $q \not\equiv 0 \bmod r$, passing requires a glitch in $(x, x + q]$, which happens for at most a $kq/N \le k/\sqrt{N}$ fraction of x. Multiples of r also pass, so the smallest passing candidate is r.
- **Cost.** Each sample needs $2L + 1$ candidates at $\mathrm{poly}(\log N)$ classical cost each. If $S_c(L) \ge s_0$ for every informative c, the expected number of samples is $O(r/(\varphi(r)s_0)) = O(\log\log r/s_0)$. With $L = k$ and Conjecture 7 this is the poster's "cost scales like the number of glitches".

What is missing: a lower bound on the mass near *informative* c, which a bound uniform in c would supply; a careful analysis of how the classical test rejects wrong candidates, including inserted symbols; and the $r \nmid N$ case, where candidates must satisfy $|y'/N - c/r| \le 1/(2r^2)$. The general-$k$ averaged guarantees in Theorem 3, Proposition 4 and Theorem 5 do not supply the informative-outcome bound.

Numerical evidence (numerical evidence only). The command `glitchy.py recover` computes the *exact* probability that one run outputs r, with $L = k$ and $\tau = 1/4$. It sums $P_{f_k}(y)$ over every outcome y for which the procedure returns r, and it applies the shift test to all x instead of a sample. Inserted symbols are new. Table 3 reports the worst placements found by a local search over positions and signs; for $k = 1$ the search covers every third position with both signs. These are search results, not proved worst cases.

N	r	$\varphi(r)/r$	no edits	$k = 1$	$k = 2$	$k = 3$	$k = 4$
3000	50	0.400	0.400	0.333 (0.83)	0.342 (0.86)	0.326 (0.81)	0.334 (0.84)
3001	50	0.400	0.379–0.392	0.350 (0.92)	0.336 (0.87)	0.338 (0.87)	0.332 (0.85)
2500	47	0.979	0.925–0.959	0.836 (0.90)	0.828 (0.88)	0.811 (0.85)	0.804 (0.84)

Table 3: Exact probability that one run of the recovery procedure outputs r, for the worst literal edit placements found. In parentheses is the ratio to the glitch-free success at the same radius $L = k$; that baseline grows with L when $r \nmid N$. Placements and signs are listed in `glitchy.py check`.

Every ratio is at least 0.81, far above the factor 1/2 that Conjecture 7 would give. The success sums the mass over all informative c, so it stays well above the per-outcome minima of Table 2. The results are identical for every τ in $[0.1, 0.5]$. Smaller τ can reject r itself, as the bound $(kr + A)/(N - r)$ above predicts.

10 Open problems

1. **Main conjecture.** Prove Conjecture 7: $S^{\mathrm{cont}}(k) \ge 1/2$. Failing that, prove any constant $c_0 > 0$ at radius $O(k)$, or find a configuration below 1/2. Known so far: it holds for $k = 1$ (Theorem 10, sharp value 0.640); exhaustive Model I search confirms it for $k = 2, 3, 4$ (Table 2) and for every sign pattern with $k \le 3$ at $N = 3000$, $r = 50$ (Section 7.3); it holds at radius $\Theta(k^2)$ (Corollary 9); and

the constant cannot be improved asymptotically (Section 8.1). The step-function form Eq. (46) is the natural setting.

2. **Extremal configurations.** Which α is worst? It is $1/2$ for $k = 1, 3$, but 0.3 for $k = 2$ and 0.4 for $k = 4$ (Table 2). Do worst cases cluster their glitches, as the packed family and the minimizers $(6, 54)$, $(3, 6, 27)$ and $(2, 4, 6, 8)$ suggest? Numerical optimization over $(u_1, ..., u_k)$ in the continuum model for $k \leq 10$ is cheap.
3. **Mixed insertions and deletions.** Only the small runs of Section 7.3 exist, and there mixed signs are never worse. Prove that equal signs are always the worst case, or find a counterexample. With mixed signs the phases $\varphi_j = e^{2\pi i \alpha s_j}$ follow a ± 1 walk instead of a geometric progression.
4. **Rigorous transfer.** Complete Proposition 12 with explicit constants, including several glitches in one period, and determine the admissible range of k relative to N and r.
5. **Period recovery.** Turn Section 9 into a theorem. Determine whether averaged bounds such as Theorem 3 can be strengthened to control informative outcomes, and whether the radius–samples trade-off of Proposition 11 is real.
6. **$r \nmid N$, per outcome.** Theorem 5 proves constant total mass on the radius-k union of rounded peaks. Extend the per-outcome formula Lemma 2 to non-integer peak positions cN/r, as in Figure 6, and control the informative outcomes.
7. **Random glitches.** An average-case model, with uniformly random positions, should be much easier and may give better constants. It is a different theorem from the worst-case one.
8. **Literature.** As of October 2026 the author found no public paper, preprint or manuscript on glitchy period finding beyond the 2019 poster. In particular, nothing is available for the QIP 2026 follow-up mentioned in the research prompt. Do not rely on that follow-up as a source. It remains worth searching for related work under other names, such as robustness of period finding or of the hidden subgroup problem to oracle errors, and Fourier sampling with corrupted oracles.
9. **Other algorithms.** The poster's future work: robustness of other Fourier-sampling or hidden-subgroup algorithms to glitchy oracles.

A Reproducibility

The numerical simulator is `glitchy.py`, at the repository root. It needs `numpy`, and `matplotlib` for plots. Lean proofs and diagram sources are separate, as described below. With uv the dependencies install automatically:

```
uv run glitchy.py check                                    # every number in this note
uv run glitchy.py plot  --N 11000 --r 100 --k 10           # random placement
uv run glitchy.py plot  --N 3000 --r 50 --blocks 6 54      # given placement
uv run glitchy.py worst --N 1500 --r 50 --k 4              # all placements
uv run glitchy.py worst --N 3000 --r 50 --k 3 --signs 1 -1 1
```

`check` runs a deterministic numerical test suite. It prints (1) the Model I worst cases of Table 2, recomputed from the logged placements and by full search; (2) the sign-pattern table of Section 7.3; (3) the one-glitch constants of Theorem 10, with the closed form checked against direct summation; (4) the packed-family table of Section 8.1; (5) the radius example of Proposition 11; and (6) checks of the residue-count identity Eq. (14) and both bounds in Theorem 3 against direct Fourier measurement probabilities. Literal tests enumerate small two- and three-edit cases and use a fixed seed for larger k, including $k \geq r$, $k > N$, mixed signs, adjacent edits, repeated or new inserted values, and truncation. They also verify the worked example in Figure 3 and equality cases of Proposition 4. (7) The nondivisible-period tests check Eq. (30), the edit-pair bound Eq. (31), rounding separation, the Fejér-kernel

expectation and both cases of Theorem 5 against the exact literal-prefix Fourier distribution. They cover half-integer peaks, mixed signs, old/new inserted symbols, truncation and overlapping windows. The theorem's proof, not these finite checks, supplies the uniform guarantee.

The poster figures were cropped from `Summer_Symposium_Poster.pdf`. The explanatory figures in Section 4.1 have editable sources in `report_figures/`. Rebuild them with:

```
mmdc -i report_figures/general_k_proof.md -o report_figures/general_k_proof.svg
python3 report_figures/build_general_k_example.py
typst compile glitchy_period_finding_report.typ
```

The Mermaid command emits `general_k_proof-1.svg` from the single fenced diagram. The committed SVGs are sufficient to compile the report; Mermaid is needed only to regenerate its proof map.

Lean verification. With elan installed:

```
cd formal
lake exe cache get       # first use: download precompiled mathlib dependencies
lake build
lake env lean Audit.lean
```

The toolchain and dependencies are pinned in `formal/lean-toolchain`, `formal/lakefile.toml`, and `formal/lake-manifest.json`. The axiom audit covers the literal-edit theorem, both sharper bounds, their neighbourhood and finite-size corollaries, the Model I bound and sharpness, the worked example, and the complete nondivisible-period matching/Fejér argument in Theorem 5. If macOS rejects the cache executable, `formal/README.md` documents an interpreter-based cache-fetch command. The verification itself uses the same Lean kernel in either case.